Worksheet 1.3 — Solving by Substitution

Grade 10 Academic (MPM2D) · Unit 1: Linear Systems

The Big Ideas

Substitution solves a system algebraically: isolate one variable, then replace it in the other equation.

- **Isolate** one variable (easiest when a coefficient is 1).
- **Substitute** that expression into the *other* equation and solve for one variable.
- **Back-substitute** to find the other variable; write the answer as (x, y).
- A true statement like $2 = 2$ means **infinitely many** solutions; a false one like $0 = 3$ means **none**.

Concept Spotlight

The idea

Substitution turns two equations into one: solve one equation for a variable, then replace that variable in the other equation, leaving a single equation in one unknown.

The steps

Isolate x or y (whichever is easiest), substitute the expression into the second equation, solve it, then back-substitute to find the other variable.

When to use it

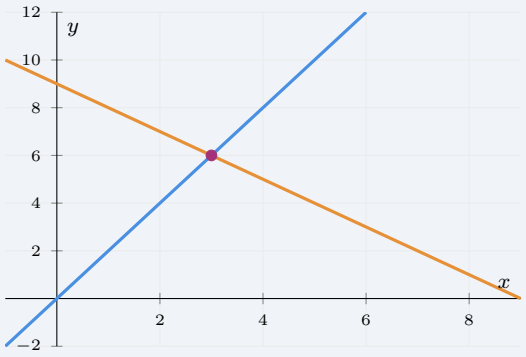
Substitution shines when one variable already has a coefficient of 1, so isolating it introduces no fractions.

Worked Examples

Example 1 — A variable already isolated

Solve: $y = 2x$ and $x + y = 9$.

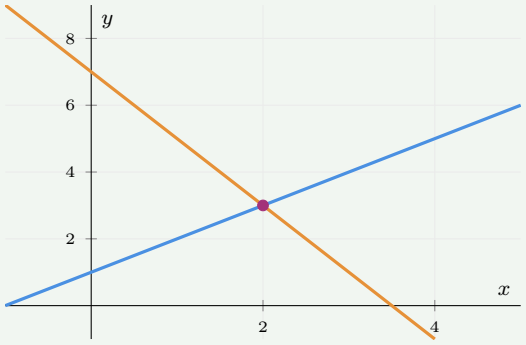
Solution. Substitute $y = 2x$: $x + 2x = 9 \Rightarrow 3x = 9 \Rightarrow x = 3$, so $y = 6$. Solution $(3, 6)$. The graph confirms it: the two lines cross exactly at $(3, 6)$.



Example 2 — Substitute and simplify

Solve: $y = x + 1$ and $2x + y = 7$.

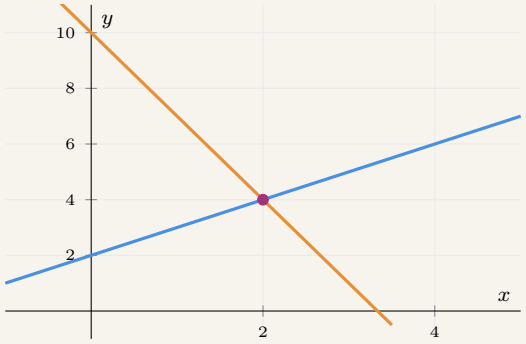
Solution. $2x + (x + 1) = 7 \Rightarrow 3x + 1 = 7 \Rightarrow x = 2$, $y = 3$. Solution $(2, 3)$. Graphing $y = x + 1$ and $y = -2x + 7$ (the second equation solved for y) shows the lines crossing at $(2, 3)$.



Example 3 — Isolating from $x =$

Solve: $x = y - 2$ and $3x + y = 10$.

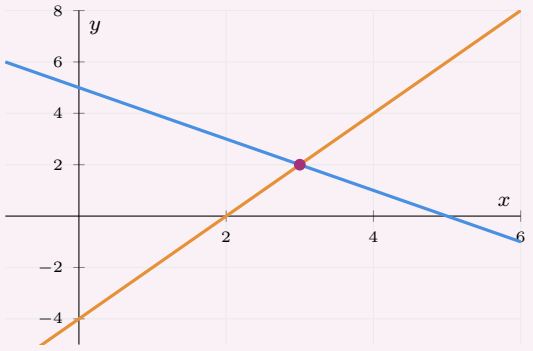
Solution. $3(y-2)+y = 10 \Rightarrow 4y-6 = 10 \Rightarrow y = 4$, $x = 2$. Solution $(2, 4)$. Rewriting both as $y = x+2$ and $y = -3x + 10$, the graph shows them meeting at $(2, 4)$.



Example 4 — Isolate first

Solve: $x + y = 5$ and $2x - y = 4$.

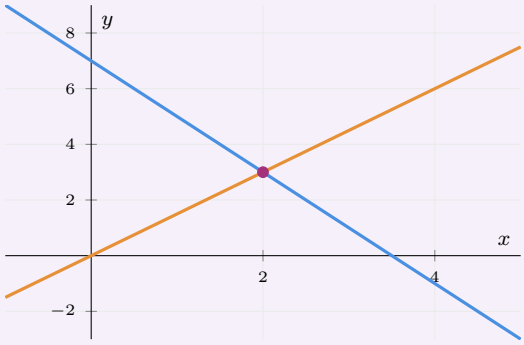
Solution. From the first, $y = 5 - x$. Then $2x - (5 - x) = 4 \Rightarrow 3x - 5 = 4 \Rightarrow x = 3$, $y = 2$. Solution $(3, 2)$. The graph of $y = 5 - x$ and $y = 2x - 4$ confirms the crossing point.



Example 5 — Larger coefficients

Solve: $2x + y = 7$ and $3x - 2y = 0$.

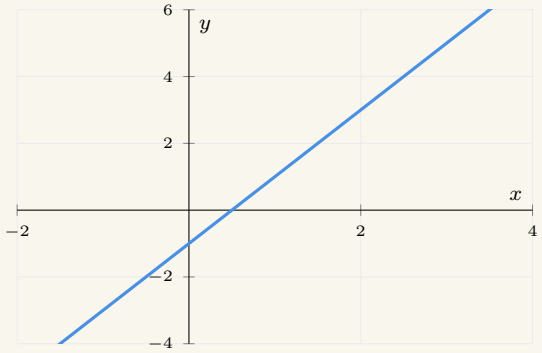
Solution. $y = 7 - 2x$. Then $3x - 2(7 - 2x) = 0 \Rightarrow 3x - 14 + 4x = 0 \Rightarrow 7x = 14 \Rightarrow x = 2$, $y = 3$. Solution $(2, 3)$. Graphing $y = 7 - 2x$ and $y = 1.5x$ (the second equation solved for y) shows the lines crossing at $(2, 3)$.



Example 6 — Infinitely many

Solve: $y = 2x - 1$ and $4x - 2y = 2$.

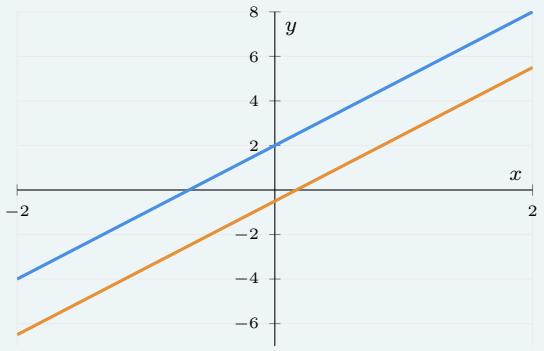
Solution. $4x - 2(2x - 1) = 2 \Rightarrow 4x - 4x + 2 = 2 \Rightarrow 2 = 2$. Always true $\Rightarrow$ **infinitely many solutions**. Graphing $4x - 2y = 2$ as $y = 2x - 1$ shows it is the exact same line — every point on it is a solution.



Example 7 — No solution

Solve: $y = 3x + 2$ and $6x - 2y = 1$.

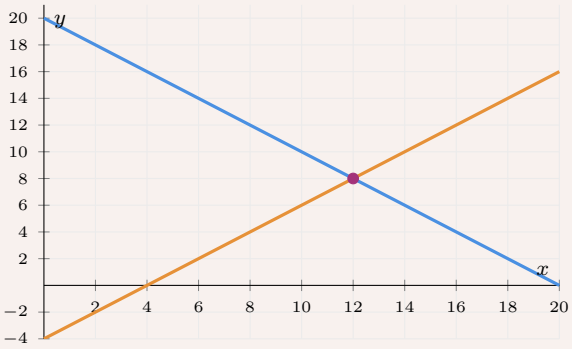
Solution. $6x-2(3x+2) = 1 \Rightarrow 6x-6x-4 = 1 \Rightarrow -4 = 1$. False $\Rightarrow$ **no solution**. Graphing $6x-2y = 1$ as $y = 3x - 0.5$ shows why: both lines have slope 3 but different y-intercepts (2 and -0.5), so they are parallel and never cross.



Example 8 — Word problem

Two numbers have a sum of 20 and a difference of 4. Find them.

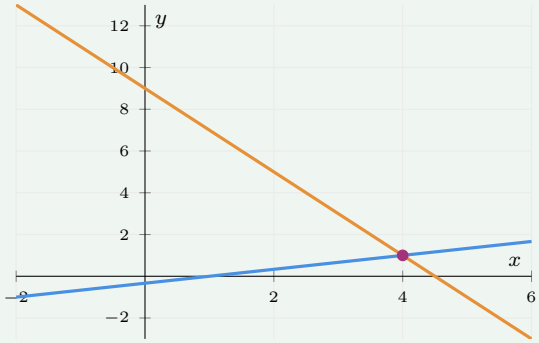
Solution. $x+y = 20$, $x-y = 4$. From the second, $x = y+4$; then $(y+4)+y = 20 \Rightarrow 2y = 16 \Rightarrow y = 8$, $x = 12$. The numbers are 12 and 8. Graphing $y = 20 - x$ and $y = x - 4$ shows the lines crossing at $(12, 8)$, matching the numbers found.



Example 9 — Multi-step with check (harder)

Solve: $x - 3y = 1$ and $2x + y = 9$.

Solution. $x = 1 + 3y$. Then $2(1 + 3y) + y = 9 \Rightarrow 2 + 7y = 9 \Rightarrow y = 1$, $x = 4$. Check: $4 - 3(1) = 1$ ✓, $2(4) + 1 = 9$ ✓. Solution $(4, 1)$. Graphing $y = \frac{x-1}{3}$ and $y = 9 - 2x$ confirms the crossing point.



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Solve by substitution: $y = 3x$ and $x + y = 8$.

Question 2

Solve by substitution: $y = x - 3$ and $x + y = 7$.

Question 3

Solve by substitution: $x = 2y$ and $3x - y = 10$.

Question 4

Solve by substitution: $x + y = 10$ and $y = x + 2$.

Question 5

Solve by substitution: $2x + y = 5$ and $y = x - 1$.

Question 6

Solve by substitution: $x - y = 3$ and $2x + y = 9$.

Question 7

Solve by substitution: $y = 2x + 1$ and $3x + y = 11$.

Question 8

Solve: $y = 4 - x$ and $2x + 2y = 8$. How many solutions?

Question 9

Solve: $y = 2x$ and $4x - 2y = 3$. How many solutions?

Question 10

Two numbers have a sum of 30, and one is twice the other. Find them.

Question 11

Solve by substitution: $3x - y = 7$ and $y = x + 1$.

Question 12

Solve by substitution: $x + 2y = 8$ and $x = y + 2$.

Question 13 — Challenge

A jar of 12 coins (dimes and quarters) is worth \$2.10. How many of each?

Question Recap & Answer Key

Question Recap

1. Solve by substitution: $y = 3x$ and $x + y = 8$.
2. Solve by substitution: $y = x - 3$ and $x + y = 7$.
3. Solve by substitution: $x = 2y$ and $3x - y = 10$.
4. Solve by substitution: $x + y = 10$ and $y = x + 2$.
5. Solve by substitution: $2x + y = 5$ and $y = x - 1$.
6. Solve by substitution: $x - y = 3$ and $2x + y = 9$.
7. Solve by substitution: $y = 2x + 1$ and $3x + y = 11$.
8. Solve: $y = 4 - x$ and $2x + 2y = 8$. How many solutions?
9. Solve: $y = 2x$ and $4x - 2y = 3$. How many solutions?
10. Two numbers have a sum of 30, and one is twice the other. Find them.
11. Solve by substitution: $3x - y = 7$ and $y = x + 1$.
12. Solve by substitution: $x + 2y = 8$ and $x = y + 2$.
13. A jar of 12 coins (dimes and quarters) is worth \$2.10. How many of each?

✓ Answer Key

1. $x + 3x = 8 \Rightarrow x = 2$, $y = 6$: $(2, 6)$.
2. $2x - 3 = 7 \Rightarrow x = 5$, $y = 2$: $(5, 2)$.
3. $6y - y = 10 \Rightarrow y = 2$, $x = 4$: $(4, 2)$.
4. $2x + 2 = 10 \Rightarrow x = 4$, $y = 6$: $(4, 6)$.
5. $2x + x - 1 = 5 \Rightarrow x = 2$, $y = 1$: $(2, 1)$.
6. $x = y + 3$; $2(y + 3) + y = 9 \Rightarrow y = 1$, $x = 4$: $(4, 1)$.
7. $5x + 1 = 11 \Rightarrow x = 2$, $y = 5$: $(2, 5)$.
8. $2x + 2(4 - x) = 8 \Rightarrow 8 = 8$: infinitely many.
9. $4x - 4x = 3 \Rightarrow 0 = 3$: no solution.
10. $x = 2y$, $x + y = 30 \Rightarrow 3y = 30 \Rightarrow y = 10$, $x = 20$.
11. $3x - (x + 1) = 7 \Rightarrow x = 4$, $y = 5$: $(4, 5)$.
12. $(y + 2) + 2y = 8 \Rightarrow y = 2$, $x = 4$: $(4, 2)$.
13. $d + q = 12$, $0.10d + 0.25q = 2.10 \Rightarrow 0.15q = 0.90 \Rightarrow q = 6$, $d = 6$.